THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A convex mirror is used as a rear-view mirror. Reason (R): It always forms virtual, erect and diminished images with a wide field of view. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A concave mirror forms a real image of a candle on a screen kept at 45 cm. If the candle is 15 cm from the mirror, the focal length of the mirror (magnitude) is 1

(a) 30 cm **(b)** 15 cm **(c)** 11.25 cm **(d)** 22.5 cm

3. An object is placed 24 cm from a lens and a real image forms at 24 cm. The lens is 1

(a) convex, f = 12 cm **(b)** concave, f = -24 cm
(c) concave, f = -12 cm **(d)** convex, f = 24 cm

4. Assertion (A): A concave lens can never form a real image. Reason (R): A concave lens diverges light rays. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A concave mirror forms a virtual image of magnification +2. The object lies at 20 cm. The focal length is 1

(a) -40 cm **(b)** -80 cm **(c)** -10 cm **(d)** -20 cm

6. When the object distance in a convex lens is made 3 times f, the image distance is 1

(a) 0.75 f **(b)** 3 f **(c)** 1.5 f **(d)** 2 f

7. Assertion (A): The image formed by a plane mirror is real. Reason (R): Plane mirror forms an image behind the mirror. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. For an object between P and F of a concave mirror, as the object moves from F towards P, the image size 1

(a) increases **(b)** decreases
(c) becomes zero first **(d)** stays the same

9. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles. 1
Reason (R): A convex mirror always forms an erect, diminished image and gives a wider field of view.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A ray of light passing through the centre of curvature of a concave mirror retraces its path after reflection. 1

Reason (R): The radius of curvature of a spherical mirror is twice its focal length.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The image of an object formed by a concave mirror is real, inverted and 1/3 of the object size at 15 cm from the mirror. Find the object distance and focal length. 2

12. A student uses a concave mirror to focus the Sun's rays on paper kept at 20 cm. Find the radius of curvature of the mirror. 2

13. Find the lateral shift reasoning: why is it zero for normal incidence? 2

14. A convex lens and a concave mirror have the same focal length of 20 cm. Compare the image distance for an object at 40 cm in each case. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A concave mirror forms a real image three times larger than the object. When the object is moved 10 cm farther away, the image becomes the same size as the object. Find f. 3

16. A student wants a 5 times magnified real image on a screen using a convex lens of f = 12 cm. Find the object distance and the screen distance. 3

17. A ray of light enters from glass (n = 1.5) to water (n = 1.33). Does it bend towards the normal? Find the relative refractive index. 3

OR

A convex lens forms a real image of a candle on a screen 80 cm away. The magnification is -3. Find u and f. 3

18. A shopkeeper uses a convex mirror to see the whole shop. A customer is 4 m from the mirror of f = 2 m. Find the image position and size ratio. 3

OR

A ray falls normally on a plane mirror. Find the angle of incidence and reflection. Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A student places a 3 cm tall object on the principal axis of a concave mirror of radius of curvature 40 cm, at a distance of 30 cm from the pole. 5

(i) Find the focal length of the mirror.

(ii) Find the position of the image.

(iii) Find the magnification.

(iv) Find the height of the image and state its nature.

20. A convex lens of f = 15 cm forms a real image on a screen. If the screen is at 45 cm from the lens find u, m and the image size for an object 6 cm tall. Then find the screen position if the object is moved to 45 cm. 5

OR

Light travels from a medium A where its speed is 2.4×10^8 m/s to a medium B where its speed is 1.5×10^8 m/s. Find n of A and B and the refractive index of B relative to A, and state how the ray bends. 5

END OF QUESTION PAPER — Total: 40 Marks